Exercise Sheet 7

Throughout, a torus T has Lie algebra $\mathfrak{t}$ and $\Lambda = \ker \exp$. Characters are $\chi_\ell(\exp H) = e^{i\ell(H)}$ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$, and $H_0 = \operatorname{diag}(i, -i) \in \mathfrak{su}(2)$. The module V_m of $\mathrm{SU}(2)$ has dimension $m+1$ and a basis $v_m, v_{m-2}, \dots, v_{-m}$ with $\Pi_m(\operatorname{diag}(w, w^{-1}))v_j = w^j v_j$.

Exercise 7.1 (Cocharacters). Let T be a torus. For $H \in \Lambda$ let $\lambda_H(e^{i\theta}) = \exp(\theta H/2\pi)$.

(a) Show that λ_H is a well-defined continuous homomorphism $\mathrm{U}(1) \to T$, and that $\lambda_{H+H'} = \lambda_H \lambda_{H'}$.

(b) Let $\lambda\colon \mathrm{U}(1) \to T$ be a continuous homomorphism. Show that $\lambda = \lambda_H$ for exactly one $H \in \Lambda$.

(c) Let $T \subset \mathrm{U}(2)$ be the diagonal torus, $\ell(\operatorname{diag}(i\theta_1, i\theta_2)) = 3\theta_1 - \theta_2$ and $H = 2\pi i \operatorname{diag}(1, 2)$. Compute $\langle \ell, H \rangle = \ell(H)/2\pi$ and compare it with $\chi_\ell \circ \lambda_H$.

Exercise 7.2 ($\mathrm{SO}(3)$ from its own torus). In $\mathfrak{so}(3)$ let

$$L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad L_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \qquad L_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

In the notation $J_i v = e_i \times v$ for $v \in \mathbb{R}^3$, these matrices are $L_x = J_1$, $L_y = J_2$ and $L_z = J_3$. Let $T = \{\exp(\theta L_z) : \theta \in \mathbb{R}\}$. Take as given that T is a maximal torus of $\mathrm{SO}(3)$ with $\mathfrak{t} = \mathbb{R}L_z$.

(a) Compute $\exp(\theta L_z)$ and Λ.

(b) Verify $[L_z, L_x] = L_y$ and $[L_z, L_y] = -L_x$ by matrix multiplication. Deduce that $\alpha(\theta L_z) = \theta$ is a root with root space $\mathbb{C}(L_x - iL_y)$. Find H_α and $Q^\vee$.

(c) Compute $X^*(T)$, Q, P, $P^\vee$, $Z(\mathrm{SO}(3))$ and $\pi_1(\mathrm{SO}(3))$.

(d) The double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ may be chosen so that its differential sends H_0 to $2L_z$. Such a cover is the double cover of Lecture 4 followed by conjugation by a fixed element of $\mathrm{SO}(3)$. Check that your answers agree with Lecture 7, which used H_0.

Exercise 7.3 (The central constraint for $\mathrm{U}(2)$). Let $\Psi\colon \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2)$, $\Psi(z, g) = zg$. It is onto with kernel $F = \{(1, I), (-1, -I)\}$. For $k \in \mathbb{Z}$ and $m \geq 0$ let $\Pi_{k,m}(z, g) = z^k \Pi_m(g)$ on V_m.

(a) Show that Ψ induces an isomorphism of Lie groups $(\mathrm{U}(1) \times \mathrm{SU}(2))/F \to \mathrm{U}(2)$. Show that $\Pi_{k,m}$ descends to $\mathrm{U}(2)$ if and only if $k \equiv m \pmod 2$.

(b) Let Π be an irreducible representation of $\mathrm{U}(2)$. Show that $\Pi \circ \Psi \cong \Pi_{k,m}$ for exactly one pair (k, m), and that $k \equiv m \pmod 2$.

(c) For $k \equiv m \pmod 2$, compute the eigenvalue of $\Pi(\operatorname{diag}(z_1, z_2))$ on v_j, and the highest weight in $\mathbb{Z}^2$.

(d) Find (k, m) for $\mathbb{C}^2$, its dual, $\det$ and $\operatorname{Sym}^2\mathbb{C}^2$. Decompose $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Exercise 7.4 (Central characters of $\mathrm{SU}(3)$). Let T be the diagonal torus of $\mathrm{SU}(3)$, with $\mathfrak{t}$, Λ and $\ell_m(H) = \sum_j m_j\theta_j$ as in Lecture 7. Use the simple roots α_{12}, α_{23}. Let $\omega = e^{2\pi i/3}$ and $H_\omega = \operatorname{diag}(2\pi i/3, 2\pi i/3, -4\pi i/3)$. For integers $a, b \geq 0$ let $V(a,b)$ be the irreducible representation with highest weight $\ell_{(a+b,b,0)}$. By the Weyl dimension formula,

$$\dim V(a,b) = \frac{(a+1)(b+1)(a+b+2)}{2}.$$

(a) Show that $P^\vee/\Lambda \cong \mathbb{Z}/3$, generated by the class of H_ω. Deduce $Z(\mathrm{SU}(3)) = \{I, \omega I, \omega^2 I\}$.

(b) Show that ωI acts on the highest weight space of $V(a,b)$ by ω^{a+2b}. Show without Schur's lemma that every weight space gives the same scalar.

(c) Tabulate $\dim V(a,b)$ and decide descent to $\mathrm{PSU}(3)$ for the six pairs

$$(a,b) = (1,0),\ (0,1),\ (2,0),\ (1,1),\ (3,0),\ (2,2).$$

(d) Show that every connected Lie group with Lie algebra $\mathfrak{su}(3)$ is isomorphic to $\mathrm{SU}(3)$ or $\mathrm{PSU}(3)$.

Exercise 7.5 (Five quotients of $\mathrm{SU}(2) \times \mathrm{SU}(2)$). Let $K_{sc} = \mathrm{SU}(2) \times \mathrm{SU}(2)$, with maximal torus the product of the diagonal tori. Identify $\mathfrak{t}$ with $\mathbb{R}^2$ by $(s,r) \mapsto (sH_0, rH_0)$. The real roots are $\pm\alpha_1, \pm\alpha_2$ with $\alpha_1(s,r) = 2s$ and $\alpha_2(s,r) = 2r$. The coroots are $H_{\alpha_1} = (1,0)$ and $H_{\alpha_2} = (0,1)$. Let $V_a \boxtimes V_b$ be $V_a \otimes V_b$ with (g,h) acting by $\Pi_a(g) \otimes \Pi_b(h)$. You may use that each $V_a \boxtimes V_b$ is irreducible, and that every irreducible representation of K_{sc} is isomorphic to exactly one of them (highest weight classification for $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, with Lecture 4).

(a) Compute Λ_{sc}, $Q^\vee$, $P^\vee$ and $Z(K_{sc})$.

(b) For each of the five subgroups $C \subset Z(K_{sc})$, compute Λ_C, the centre of K_{sc}/C and $\pi_1(K_{sc}/C)$.

(c) For each C, decide which $V_a \boxtimes V_b$ descend to K_{sc}/C. Check the answer against $X^*(T_C)$.

(d) Show that the quotients by $\langle(-I, I)\rangle$ and $\langle(I, -I)\rangle$ are isomorphic. Show that the quotient by $\langle(-I, -I)\rangle$ is not isomorphic to them. How many isomorphism classes do the five quotients form?

Exercise 7.6 ($\mathrm{U}(2)$ versus $\mathrm{U}(1) \times \mathrm{SO}(3)$). Let $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be the double cover of Lecture 4, with kernel $\{\pm I\}$, and let $K_2 = \mathrm{U}(1) \times \mathrm{SO}(3)$. For $k \in \mathbb{Z}$ and $m \geq 0$ let $\Pi_{k,m}(z,g) = z^k \Pi_m(g)$ on V_m.

(a) Show that $\mathrm{U}(2)$ and K_2 have isomorphic Lie algebras and isomorphic centres.

(b) Show that $(z,g) \mapsto (z, \rho(g))$ identifies K_2 with $(\mathrm{U}(1) \times \mathrm{SU}(2))/F'$, where $F' = \{(1,I), (1,-I)\}$. Which $\Pi_{k,m}$ descend to K_2?

(c) Use $\pi_1(K) \cong \Lambda/Q^\vee$ to show that $\pi_1(K_2) \cong \mathbb{Z} \oplus \mathbb{Z}/2$. Conclude that $K_2 \not\cong \mathrm{U}(2)$.

(d) (Optional extension.) Show that $(z,g) \mapsto (z^2, \rho(g))$ induces an isomorphism $(\mathrm{U}(1) \times \mathrm{SU}(2))/F'' \cong K_2$, where $F'' = \{(\pm 1, \pm I)\}$. Explain why this is not the presentation of the global structure theorem.